

Introduction

This project is an extension of CW2 with the addition of considering battery consumption, hovering speed, scheduling system and priority dispatches. In the ideal situation the system operates as a continuously running REST service which would accept input of the drones' metadata and streams of delivery dispatch requests.

This design intends to simulate realistic scenarios in computing flight paths whilst respecting:

- Drones must avoid restricted zones
- Drones should manage battery properly
- Choose the best option between multiple drones
- Schedule deliveries
- Logic that prioritises urgent jobs differently from normal job

The system maintains a time evolving state which includes the current availability of each drone, its remaining battery, location and charging status. Upon receiving delivery request the planner updates this state and compute assignment accordingly

Intended Audience

This system is designed for organisations that requires reliable drone-based delivery logic such as hospitals, pharmacies that require transportation of medical/ equipment dispatches. This also apply to delivery companies who are trying to integration drone delivery into their business model. Individuals who are working towards building their own routing/ scheduling system could also use the system to test ideas/ check their work.

In general this system is useful for anyone who needs a system that plans routes/paths, manage battery and scheduling

System Architecture

A* Pathfinding

The system uses A* pathfinding where lnglat coordinates are converted into grid cells where each grid step represents a step taken by the drone. The cost per step are computed based on the drone's payload and battery efficiency. Each node in the tree stores the drone's location, remaining battery energy and the accumulated time elapsed. The g-cost reflects the energy spent and time taken whereas the f cost combines g cost and a heuristic which estimates the remaining distance to the target. These are also data stored in each node.

A full mission consists of an outbound path (service point to delivery point) and inbound bath (delivery point to service point). The system verifies that both halves of the route satisfies battery and capability constraints

Battery Consumption Model

Battery model introduces three dynamics:

- Per step consumption – Each step incurs base energy burn with an increase with heavier payload
- Degradation – Repeated mission increases long term battery cost

- Idle charging - Drones are charging when idling at a service point with the rate of charging being different at each service point.

The system has two types of charging, one being passive charging (automatic when the drone is not assigned to a delivery) and active charging (required when the next delivery estimated battery used is greater than current battery level). Active charging time are calculated into the delivery time given the battery deficit and charging rate.

Scheduling

Scheduler manages the resource allocation across multiple drones. For each unassigned delivery, the scheduler evaluates each drone independently by checking:

- Earliest time which a drone can start the delivery
- Amount of active charge needed to reach the min charge required
- Expected delivery time and whether that fits the drone's availability
- Drone's capability constraints
- Respecting priority modes - **NORMAL** → choose the most battery efficient drone **IMPORTANT** → choose fastest completion time

Each planning iteration constructs a set of feasible options and one assignment is chosen which triggers execution. The process is repeated until the queue is empty (all delivery request completed/ leftover deliveries that can't be executed)

Limitations & Possible Extensions

Given more time, there are some useful and practical extensions that could be done to improve the quality of this system

- weather (wind speed/direction) - By taking into wind direction we can better simulate the real world where going against the wind would decrease the speed and increase battery consumption
- Better queuing - queueing is currently modelled as FCFS but can become priority-based where more important deliveries are considered first
- Better pathfinding - A* uses fixed step size but with adaptive step size could accelerate search and reduce planning time
- battery chemistry ageing model could be more realistic
- probabilistic delays (stochastic events) not integrated
- Collision – With increasing number of drones and delivery in the real world we have to ensure drones do not fly through each other